

Sensitivity analysis of GRNmap and new features for GRNsight: open source software for dynamical systems modeling and visualization of small-scale gene regulatory networks

Kam D. Dahlquist¹, Ben G. Fitzpatrick², John David N. Dionisio³, Ngoc K. Tran³, Nikki C. Chun^{1,2}, Cecilia J. Zaragoza³, A'Kaia L. Phelps³, Amelie T. Dinh³, Milka Y. Zekarias³

¹Department of Biology, ²Department of Mathematics, ³Department of Computer Science, Loyola Marymount University, 1 LMU Drive, Los Angeles, CA 90045 USA

Email: kdahlquist@lmu.edu

Project Websites: <http://kdahlquist.github.io/GRNmap/> and <http://dondi.github.io/GRNsight/>

Source Code: <https://github.com/kdahlquist/GRNmap/> and <https://github.com/dondi/GRNsight>

License: BSD license, <https://github.com/kdahlquist/GRNmap/blob/master/LICENSE> and <https://github.com/dondi/GRNsight/blob/master/LICENSE.txt>

A gene regulatory network (GRN) is a set of transcription factors that regulate the expression of genes encoding other transcription factors. GRNmap is an open source MATLAB package that uses ordinary differential equations to model dynamics of small-scale GRNs. The program estimates production rates, expression thresholds, and regulatory weights for each transcription factor in the network based on gene expression data and then performs forward simulations of model dynamics. While the model has been successfully used to understand networks of 15-20 genes, we wanted to closely examine how it works on a smaller scale to determine parameter sensitivity. All 21 possible “toy” networks of 3 nodes and 4 edges were created based on the simulated expression data outputted when known parameters were run through a forward simulation. Then the simulated data was used to estimate the parameters again. Comparison of the known to estimated parameters showed that estimating production rates in addition to weights and thresholds reduced the accuracy of the results. The model was also sensitive to the direction and magnitude of the weight parameters for networks with the same connectivity. GRNsight is an open-source web application for visualizing such models of GRNs, automatically displaying a graph where edges are color-coded based on the activation and repression relationships, and nodes are color-coded with time course gene expression data. Backend gene expression and gene regulatory network databases allow users to generate a network, color the nodes with expression data, and export an Excel workbook for modeling in GRNmap. <http://kdahlquist.github.io/GRNmap/>; <https://dondi.github.io/GRNsight/>